

# Divy Patel

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## Education

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| <b>Arizona State University (ASU)</b> , Tempe, AZ<br>Master of Science in Computer Science (MScS) (GPA: 3.55/4)               | Aug. 2022 – Dec. 2023 |
| <b>Pandit Deendayal Energy University (PDEU)</b> , India<br>B.Tech in Information and Communication Technology (GPA: 9.22/10) | Apr. 2018 – May 2022  |

## Technical Knowledge

**Programming Languages:** Java, Python, R, C++, C#, JavaScript, SQL (MySQL, PostgreSQL), GraphQL, HTML  
**Frameworks:** RESTful APIs (Microservices), Django, LangChain, TensorFlow, PyTorch, OpenCV, Scikit-learn  
**Tools and Technologies:** AWS (Lambda, EC2, S3, DynamoDB), Docker, CI/CD, Git, Redis, Jira, Agile, Scrum

## Work Experience

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| <b>GlobalLogic</b> (client: Walmart), Dallas, Texas<br>● Developed Python/Django RESTful APIs enabling external TV control through mobile, HDMI-CEC, and IoT devices.<br>● Integrated new hardware abstraction layer into TV OS components, removing manufacturer library dependency across chipsets.<br>● Built a Generative AI-based BDD test generation pipeline for Behave suites, cutting AI cost per module from \$300–400 to \$40–60.<br>● Migrated smartcast-platform-layer TV middleware from Python to MicroPython via AI code transformation, validated by BDD suites.<br>● Automated defect tracking and reporting, saving the team lead over 6 hours weekly.<br>● Consolidated five middleware release branches into one stable branch, eliminating cherry-pick overhead and fix divergence. | <b>Software Engineer</b>              | Oct. 2024 – Present   |
| <b>Arizona State University</b> , Tempe, AZ (part-time)<br>● Configured Linux servers and installed dependencies for the website, database, and ML needs.<br>● Assessed team requirements and recommended hardware for better performance and cost-efficiency.<br>● Contributed to the development of an LLM to generate daily tasks for artists based on their long-term goals.                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Engineer</b>                       | Jul. 2024 – Dec. 2024 |
| <b>Arizona State University</b> , Glendale, AZ (part-time)<br>● Developed programs to analyze genome data according to the lab's requirements under the supervision of Dr. Sreetharan Kanthaswamy.<br>● Contributed to a project analyzing genetic diversity and mutations across different monkey breeds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Research Technician</b>            | Apr. 2024 – Oct. 2024 |
| <b>OleOleSport</b> , Remote (Santa Monica, CA)<br>● Built a football history chatbot using LangChain, Mistral 7B, and architectures like RAG and CRAG.<br>● Experimented with models, including Llama, achieving <b>98%</b> accuracy for Arsenal team history Q&A.<br>● Designed and developed a Flutter-based front-end for a football fan community website.                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Software Engineer</b>              | Apr. 2024 – Jul. 2024 |
| <b>Quartic.ai</b> , Remote<br>● Built real-time data pipelines pre-processing live data feeds with Python, GraphQL, and Redis, deployed via CI/CD.<br>● Optimized the data pipeline feeding ML models for predictive analytics and real-time risk monitoring.<br>● Developed a live risk monitoring dashboard on pipeline outputs using Grafana and ReactJS.                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Data Engineer</b>                  | Jul. 2021 – Jun. 2022 |
| <b>HOPS Healthcare</b> , India<br>● Achieved <b>85%</b> accuracy in mobile-based skin disease detection by implementing OpenCV, TensorFlow, and CNNs.<br>● Developed a noise reduction algorithm in Python, aiding detection of heart conditions from stethoscope audio.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Artificial Intelligence Intern</b> | Apr. 2021 – Dec. 2021 |

## Projects

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| <b>LLM for Document Based Question Answering</b> <br>● Developed Large Language Model (LLM) using RAG implementation to interpret PDFs, CSV, and JSON files, with <b>98%</b> accuracy.<br>● Used LangChain for efficient storage and retrieval, ensuring accurate responses by implementing Prompt Engineering.<br>● Enhanced chat automation processes by testing model performance on diverse and complex queries. | <b>Project</b> |
| <b>Smart Parking Database Management System</b> <br>● Led development of a comprehensive database management system for smart parking lot application.<br>● Generated 24,000 dummy data entries to thoroughly test and validate the database system.<br>● Documented SQL queries, query output snapshots, and CSV exports for database testing and analysis.                                                         | <b>PDEU</b>    |
| <b>Graph Operations and Search</b><br>● Developed a Java application for graph operations with DOT parsing, node and edge management, and visualization using OOP patterns.<br>● Implemented Breadth-First Search (BFS) and Depth-First Search (DFS) with unit tests covering core operations.<br>● Provided a command-line interface for graph interaction, searches, and export to .dot and image formats.                                                                                            | <b>ASU</b>     |

## Recent Achievements

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| ● Publication in Materials Today: Proceedings on <a href="#">Artificial intelligence powered material search engine</a>                 | Apr. 2022 |
| ● Publication in American Journal of Primatology on <a href="#">Sequencing Reveals MHC Diversity in Captive Rhesus Macaque Colonies</a> | Jan. 2026 |